

Emma Lhermitte

emlhe | emlhe.github.io | emma.lhermitte@univ-brest.fr | +336.24.57.07.66

SUMMARY

PhD student in the fields of medical image analysis, deep learning, computational neurosciences. Looking for a postdoctoral position in related fields.

EDUCATION

2022 - present	University of Western Brittany , Brest — PhD <i>Cerebral morpho-structural changes in children with cerebral palsy aged 1 to 4 years after HABIT-ILE motor therapy</i> - Processed brain imaging using deep learning - Developed quantitative methods for cerebral morpho-structure analysis - Identified neural correlates of clinical assessments using statistics - Supervised masters trainees
2022	Polytech Angers , Angers — Master's degree Research master in Dynamics and Signals
2019 - 2022	Polytech Angers , Angers — Engineering degree Specialized in computer science

WORK EXPERIENCE

Teaching assistant — IMT Atlantique, Brest (FR)	Sept 2024 - present
Supervised projects for segmenting medical images using deep learning models	
Supervised Applied Statistics courses	
Teaching assistant — University of Western Brittany, Brest (FR)	Jan 2024 - May 2024
Supervised courses in networking, fundamental algorithmics and testing, web development	
Internship — LARIS, Angers (FR) and IT Carlow, Carlow (IR)	March 2022 - July 2022
Did research work in the field of texture images classification using machine learning	
Internship — THALES SIX GTS, Laval (FR)	March 2022 - July 2022
Developped a reliability software applied to radio-telecommunication electronic equipments	

PROJECTS

SynthsegStroke
[Link to Github project](#)

Paediatric brain segmentation in presence of lesion using synthetic images

RESEARCH

Publications

Lhermitte, E., Hilal, M., Furlong, R., O'brien, V., & Humeau-Heurtier, A. (2022). Deep learning and entropy-based texture features for color image classification. *Entropy*, 24(11), 1577.

Lhermitte, E., Dinomais, M., Oyaneder, R. A., Bleyenheuft, Y., Guzzetta, A., Brochard, S., & Rousseau, F. (2024). Validating stroke lesion segmentation methods using mri in children: Transferability of deep learning models. *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 1–5.

Lhermitte, E., Valabregue, R., Dinomais, M., Araneda, R., Bleyenheuft, Y., Guzzetta, A., Proisy, M., Brochard, S., & Rousseau, F. (2025). Synthetic learning: A novel approach for segmenting structures in children brains with perinatal stroke. *ArXiv preprint*.

Talks at conferences

EACD 2025 - Early anatomo-functional correlations using voxel-based lesion-symptom mapping in children with left-sided stroke lesion.

RFIAP 2025 - Validating stroke lesion segmentation methods using mri in children: transferability of deep learning models.

ISBI 2024 - Validating stroke lesion segmentation methods using mri in children: transferability of deep learning models.

SKILLS

Computing skills	Linux, Bash, Python, R, Latex, Matlab
Libraries	PyTorch, Tensorflow, TorchIO, OpenCV
Neuroimaging tools	FreeSufer, CAT12, SPM, FSL, Nipype, ANTs
Languages	French (native), English (B2), Spanish (B1)